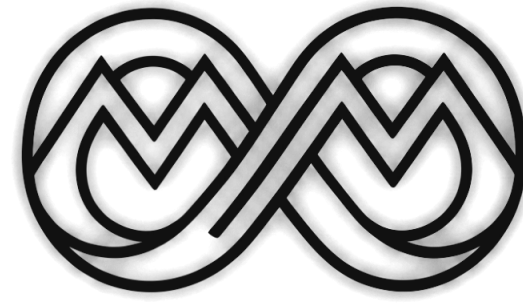




The MLOps Practitioner course

Course guide

[STRATEGY](#) · [TRACKS](#) · [CHECKLISTS](#) · [RUBRIC](#) · [TIMELINE](#)

Aya Nasser Salama · MLOps MENA Community

What is in this guide

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Six weeks, week by week

Sundays · 7:00 PM Cairo

Every live session is on a Sunday at 7:00 PM Cairo time. A mini project closes every session.

SUN · AUG 16

SUN · AUG 23

NO SESSION

SUN · SEP 6

SUN · SEP 13

SUN · SEP 20

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Week 1 · From Notebook to Production-Ready Code
Packaging, ML APIs, Docker, logging and testing — 7 lessons

Week 2 · MLOps Core — Tracking, Versioning, Automation
MLflow, DVC, GitHub Actions and Terraform — 6 lessons

Week 3 · Project Week — no lecture
Build time: apply Weeks 1–2 and revise the first two modules

Week 4 · Inference, Serving and Release Strategies
Airflow, serving stacks, load testing and rollouts — 11 lessons

Week 5 · Model Optimization — Faster, Smaller, Cheaper
Pruning, quantization, distillation and compilation — 11 lessons

Week 6 · Observability and Drift Detection
Drift methods, dashboards, LLM tracing and guardrails — 12 lessons

Foundations and automation

● WEEK 1 · AUG 15–21

From Notebook to Production-Ready Code

Live · Sun Aug 16, 7:00 PM GMT+3

- MLOps maturity model
- Python packaging and project structure
- Building ML APIs and serialization formats
- Docker and containerization
- Structured logging and testing with pytest

Mini project at the end of the session

● WEEK 2 · AUG 22–28

MLOps Core — Tracking, Versioning, Automation

Live session · Week 2

- MLflow experiment tracking
- Model registry and continuous training
- Data versioning with DVC
- CI/CD with GitHub Actions
- Infrastructure as code with Terraform

Mini project at the end of the session

○ WEEK 3 · AUG 29 – SEP 4

Project Week — no lecture

No live session

-
- Start working on your chosen project
 - Apply the principles from Weeks 1 and 2
 - Implement and revise the first two modules

Catch-up and build time

Serving, optimization, observability

● WEEK 4 · SEP 5–11

Inference, Serving and Release Strategies

Live session · Week 4

- Orchestration with Apache Airflow
- Three inference patterns: REST, batch, streaming
- Serving with FastAPI and BentoML
- GPU and CPU accelerated serving, LLM serving
- Load testing and release strategies

Mini project at the end of the session

● WEEK 5 · SEP 12–18

Model Optimization — Faster, Smaller, Cheaper

Live session · Week 5

- Pruning and quantization (PTQ, QAT)
- Knowledge distillation
- Compilation: TensorRT, ONNX Runtime, OpenVINO
- Edge deployment and LLM quantization
- Benchmarking and knowing when to stop

Mini project at the end of the session

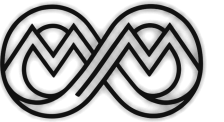
● WEEK 6 · SEP 19–25

Observability and Drift Detection

Live session · Week 6

- Why models degrade: the drift taxonomy
- Statistical detection and Evidently AI
- Concept, label and embedding drift
- Prometheus, Grafana and Langfuse tracing
- RAG evaluation, cost monitoring, guardrails

Mini project at the end of the session



Two certificates

Certifications · attendance · feedback · projects

ATTENDANCE CERTIFICATE

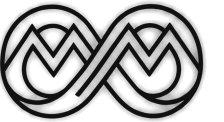
Free — awarded to everyone who completes the course

- ☐ Attend all 5 sessions (or watch the recording)
- ☐ Fill the quick feedback form after EACH session
- ☐ Fill the final course feedback form at the end
- ☐ That's it — no project required

PROJECT CERTIFICATE

Optional — awarded for completing the full project

- ☐ Complete the end-to-end project (choose 1 of 2)
- ☐ Submit your GitHub repo + README
- ☐ Review at least 1 other student's project
- ☐ Write a detailed written feedback (min. 300 words)



Attendance, and how pairing works

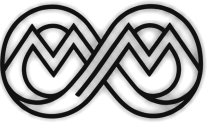
How I know you attended

After each session a short feedback form is shared in the group. It takes 2 minutes. Fill it → you're marked present. Skip it → that session doesn't count, even if you were in the room.

Peer review — project certificate only

You review 1 project from another student. I assign the pairing. Your review must cover: code quality, README clarity, one thing done well, two things to improve, and how you would extend the project.

Two project tracks — choose one. Both graded on the same 10-point rubric.



What to have in place first

You will get far more out of the five sessions if you arrive with these two things done.

1

Study the fundamentals

At minimum, the supervised learning course from Andrew Ng's Machine Learning Specialization on Coursera. You need to understand what a model is before you learn how to ship one.

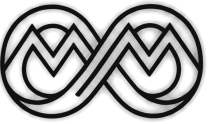
[coursera.org → Machine Learning Specialization](https://www.coursera.org/specializations/machine-learning-fundamentals)

2

Train one model yourself

Do at least one project on Kaggle that trains a simple ML model — any data type: video, images, CSVs, audio or text. The subject does not matter. Having trained something end to end does.

Both are free. Do them before Session 1, not during it.



Two projects, one standard

PROJECT 1 — DEEP LEARNING

Arabic Sentiment Analysis

PyTorch · AraBERT · TensorRT · BentoML · Triton

Best if you have a GPU and are interested in model optimization

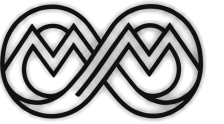
PROJECT 2 — LLM / RAG

Arabic Legal Document Q&A

vLLM · RAG · RAGAS · Langfuse · Guardrails

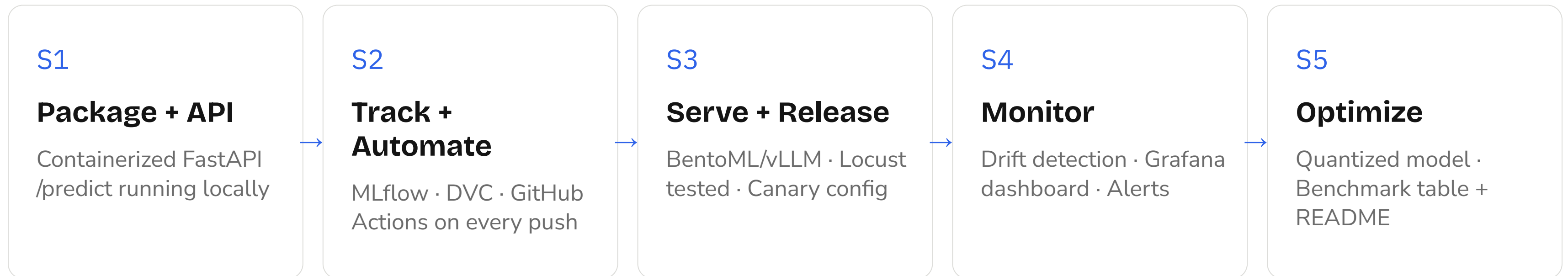
Best if no GPU — focused on evaluation and observability

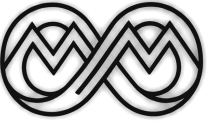
Both tracks · same 10-point rubric · same deadline · same certificate



The project strategy

One project. Five sessions. One working piece per session.





How the project runs

1 Choose once

DL or LLM — commit at Session 1 and build the same project all 5 sessions.

2 Commit per session

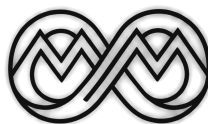
At least one GitHub commit per session. Code is your evidence of learning.

3 Same rubric

10 identical criteria. No track is easier than the other.

4 Peer review is required

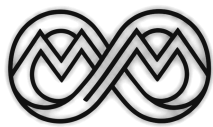
You cannot earn the project cert without reviewing another student.



The 10-point rubric

Same criteria for both tracks — no track is easier. Criteria 01–05.

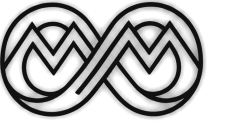
#	AREA	SESSION	WHAT YOU BUILD	PASS CONDITION
01	Code & Packaging	S1	pyproject.toml, src/ layout, OOP model class, type hints	pip install -e . works. Model loads without errors.
02	API Endpoint	S1	/predict with Pydantic, /health, async handlers	curl /predict returns correct output.
03	Docker	S1	Dockerfile, docker-compose.yml, 3-command README	docker compose up && curl localhost:8000/predict works on reviewer’s machine.
04	MLflow Tracking	S2	≥ 5 runs, params + metrics + artifacts, best model in Registry	MLflow UI shows comparison. Model promoted to Production.
05	DVC Data Versioning	S2	Dataset tracked, dvc.yaml pipeline, dvc repro works	git checkout + dvc pull + dvc repro reproduces result.



The 10-point rubric

Criteria 06–10.

#	AREA	SESSION	WHAT YOU BUILD	PASS CONDITION
06	GitHub Actions CI/CD	S2	Lint → test → build → push on every PR. Quality gate.	Merge a PR. Actions green. Docker image pushed to registry.
07	Production Serving	S3	BentoML (DL) or vLLM + BentoML (LLM). Locust tested.	p95 latency documented. Locust report in /reports/.
08	Monitoring	S4	≥ 1 drift metric (DL) OR RAGAS faithfulness (LLM). Grafana panel.	Screenshot in README. Alert threshold set.
09	Peer Review	Post S5	Written review ≥ 300 words, all 5 areas	Submitted on time. Covers: setup, code, strength, 2 improvements, extension.
10	README & Architecture	S5	3-command setup, architecture diagram, session changelog	Reviewer runs it without asking anything.



Arabic Sentiment Analysis

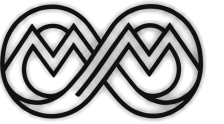
E-commerce product reviews in Arabic · PyTorch · AraBERT · BentoML · TensorRT · Triton

The business problem

An Arab e-commerce platform (Noon.com, Amazon.eg) receives 50,000 Arabic product reviews per day. They need to classify each as Positive, Negative, or Neutral — in Arabic — to power product rankings, seller ratings, and customer service prioritization.

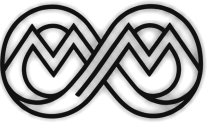
Why this model for this course

- GPU optimization and TensorRT actually matter here
- Arabic NLP — directly relevant to MENA students
- Real drift — Ramadan vocabulary, slang, dialect shifts
- All 3 inference patterns apply: web, batch, streaming
- Session 5 distillation: 12-layer → 6-layer student



What you ship per session

S1	Package + API	AraBERT wrapper, /predict: {"text": "رائع المنتج"} → {"label": "positive", "confidence": 0.94}
S2	Track + Version	MLflow: 5+ runs. DVC: labeled reviews. GitHub Actions CI on push.
S3	Serve	BentoML micro-batching. TensorRT FP16. Locust: p95 before vs after.
S4	Monitor	Evidently PSI on vocabulary. Prometheus + Grafana. Alert: PSI > 0.25.
S5	Optimize	Distil 12→6 layers. INT8 ONNX. Benchmark: accuracy × latency × size.



Completion checklist

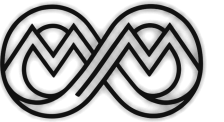
Code, packaging, tracking and versioning. Verified together with the instructor. Part 1 of 2.

CODE & PACKAGING

- ☐ src/ layout with pyproject.toml — pip install -e . works
- ☐ AraBERT or CAMEL-BERT loaded in a class with type hints
- ☐ FastAPI /predict: {text: str} → {label, confidence, model_version}
- ☐ Pydantic rejects empty text — 422 returned, tested with curl
- ☐ /health returns {status: healthy} — used by Docker healthcheck
- ☐ docker build -t arabic-sentiment . passes
- ☐ docker compose up starts service on port 8000
- ☐ README: exactly 3 commands to run on any machine

EXPERIMENT TRACKING & DATA VERSIONING

- ☐ MLflow tracking server running — UI accessible
- ☐ ≥ 5 runs: model_name, lr, batch_size, accuracy, f1_macro
- ☐ MLflow comparison screenshot in /reports/mlflow_comparison.png
- ☐ Best model registered as 'ArabicSentiment' → promoted to Production
- ☐ Labeled dataset tracked with DVC — dvc status shows clean
- ☐ dvc repro reproduces training with same metrics (± 0.005)
- ☐ GitHub Actions: lint → test → docker build → push on main
- ☐ CI fails if f1_macro drops below baseline — documented in README



Completion checklist

Serving, monitoring and optimization. Part 2 of 2.

SERVING & RELEASE

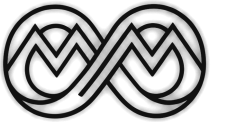
- ☐ BentoML service with batchable=True Runner — bentoml models list shows model
- ☐ bentoml serve runs /predict at localhost:8000
- ☐ Locust report before optimization in /reports/locust_fastapi.html
- ☐ TensorRT FP16 engine built — committed or documented
- ☐ Locust after TensorRT in /reports/locust_trt.html — p95 in README
- ☐ Batch scoring: $\geq 1,000$ samples, output to /data/scoring/output/
- ☐ nginx canary config: weight=5/95 — rollout stages in README

MONITORING & OBSERVABILITY

- ☐ Evidently DataDrift report on latest batch — saved in /reports/
- ☐ PSI computed for text_length and confidence_score features
- ☐ Drift check triggered after every batch scoring run
- ☐ Prometheus exposes PSI score at /metrics
- ☐ Grafana: ≥ 2 panels (p95 + PSI) — screenshot in README
- ☐ Alert: $\text{PSI} > 0.25 \rightarrow$ notification documented in README

MODEL OPTIMIZATION

- ☐ 12-layer AraBERT distilled to 6-layer student — model file committed
- ☐ INT8 ONNX quantization applied — model_int8.onnx committed
- ☐ Benchmark table: original vs distilled vs INT8 (accuracy, p95, size)
- ☐ Final architecture diagram in README covering all 5 sessions



Arabic Legal Document Q&A

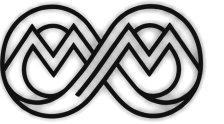
Arabic legal documents · Embeddings + Vector DB · vLLM · RAGAS · Langfuse · Guardrails

The business problem

A law firm or government authority in the Arab world needs a system that answers questions about Arabic legal documents — contracts, Egyptian Civil Code articles, Saudi regulations — accurately and with citations. Hallucinations are legally unacceptable.

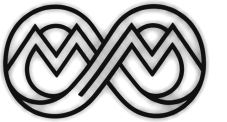
Why this project for this course

- RAGAS evaluation is genuinely hard — deep S4 content
- No GPU required to start — API-based LLM works fine
- Embedding drift is real — legal language shifts across jurisdictions
- Langfuse tracing reveals the full RAG chain
- AWQ quantization in S5 makes the LLM run fully offline



What you ship per session

S1	Package + API	src/rag.py (ingest+query), /ask: {"question": "ما هي شروط العقد؟"} → {"answer": "...", "sources": [...]}
S2	Track + Version	MLflow: chunking experiments (chunk_size, overlap, RAGAS faithfulness). DVC: document corpus.
S3	Serve	BentoML wraps RAG. vLLM serves generative model. Streaming /ask.
S4	Monitor	RAGAS on ≥ 50 questions. Langfuse traces every request. Cosine drift. Token cost.
S5	Optimize	AWQ-4bit generative model. Re-ranker distillation. RAGAS before vs after.



Completion checklist

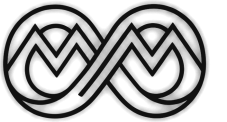
Code, packaging, tracking and versioning. Verified together with the instructor. Part 1 of 2.

CODE & PACKAGING

- ☐ src/ layout with pyproject.toml — pip install -e . works
- ☐ Ingestion pipeline: PDF/text → chunk → embed → vector store
- ☐ FastAPI /ask: {question: str} → {answer: str, sources: list[str]}
- ☐ Pydantic rejects empty question — 422 returned and tested
- ☐ /health returns {status: healthy, documents_indexed: N}
- ☐ Dockerfile includes vector store and embedded documents
- ☐ docker compose up starts service on port 8000
- ☐ README: 3 commands to run Q&A on any machine

EXPERIMENT TRACKING & DATA VERSIONING

- ☐ MLflow logs each config: chunk_size, overlap, embedding_model, faithfulness
- ☐ ≥ 5 runs compared — MLflow screenshot in /reports/
- ☐ Best chunking config registered in MLflow Registry
- ☐ Document corpus tracked with DVC — dvc pull fetches all source docs
- ☐ dvc repro re-indexes corpus reproducibly from tracked documents
- ☐ GitHub Actions: lint → test → rebuild index → push Docker image
- ☐ CI fails if RAGAS faithfulness < 0.75 on 20-question test set



Completion checklist

Serving, monitoring and optimization. Part 2 of 2.

SERVING & RELEASE

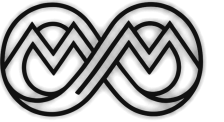
- ☐ BentoML wraps RAG pipeline with async /ask endpoint
- ☐ vLLM serves the generative LLM — model name in README
- ☐ Streaming: tokens appear progressively in curl output
- ☐ Locust report at 50 concurrent users in /reports/
- ☐ Batch re-indexing script tested with at least one new document
- ☐ Canary rollout config documented in README

MONITORING & OBSERVABILITY

- ☐ RAGAS on ≥ 50 questions — all 4 metrics logged
- ☐ RAGAS results stored in MLflow — trend visible across sessions
- ☐ Alert: faithfulness $< 0.80 \rightarrow$ notification triggered
- ☐ Langfuse self-hosted — every /ask creates a trace with spans
- ☐ RAGAS faithfulness score attached to each Langfuse trace
- ☐ Cosine similarity on query embeddings vs baseline — drift logged
- ☐ Token usage in Prometheus — Grafana shows cost/hour
- ☐ Guardrails PII detection active on all /ask responses

MODEL OPTIMIZATION

- ☐ Generative model quantized to AWQ-4bit — weights documented
- ☐ RAGAS scores: original vs quantized — ≤ 0.03 faithfulness drop
- ☐ Latency before vs after quantization in README
- ☐ Final architecture diagram covering all 5 session components



Peer review

Required for the project certificate — you cannot earn it without reviewing another student.

AS THE REVIEWER — WHAT YOU SUBMIT

- ☐ Cloned the assigned student's repo and followed the README
- ☐ Service started — documented any issues encountered
- ☐ Written review ≥ 300 words as a GitHub Issue on their repo
- ☐ PDF copy of the review sent to the instructor
- ☐ Section 1 — Setup: did it run? how long? what broke?
- ☐ Section 2 — Code Quality: 2 specific patterns (good + refactor)
- ☐ Section 3 — One Genuine Strength: specific, production context
- ☐ Section 4 — Two Specific Improvements: concrete and actionable
- ☐ Section 5 — Extension Idea: what, why, how — one paragraph

AS THE REVIEWEE — WHAT YOU RESPOND

- ☐ Read the review carefully before responding
- ☐ One GitHub comment per review section
- ☐ Each improvement: implement it or explain why you disagree
- ☐ Extension idea: acknowledge and note if you plan to implement it
- ☐ Response submitted within 3 days of receiving the review

Peer review pairing is assigned by the instructor after Session 5.

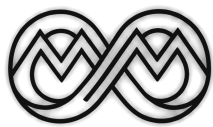


How to write a good review

Minimum 300 words. All 5 areas. I read every review — quality matters.

1	Can I run it?	Clone. Follow README. Does it work? How long? What broke?	Document every friction point. If it ran fine, note what was clear.
2	Code Quality	Type hints? OOP? Model loaded once? Repeated code?	Name 2 patterns: one that's good, one to refactor. Be specific.
3	One Genuine Strength	The ONE thing done notably well. 'Good code' is not useful.	One paragraph. What impressed you and why it matters in production.
4	Two Improvements	Not 'add more tests'. Specific enough to act on immediately.	Example: 'Locust wait_time should be between(0.5, 2.0) for realism'
5	Extension Idea	One concrete thing to build next to extend scope or production-readiness.	What it is, why it matters, how you'd approach it — one paragraph.

Submit as a GitHub Issue AND send a PDF to the instructor · Student responds with one comment per point



Timeline

During the course — one commit per session.

End of Session 1	GitHub repo created. /predict running in Docker. 3-command README.
End of Session 2	≥ 5 MLflow runs logged. DVC pipeline clean. GitHub Actions green.
End of Session 3	BentoML or vLLM service running. Locust report submitted.
End of Session 4	Drift detection or RAGAS running. Grafana dashboard live.
End of Session 5	Optimization complete. Benchmark table. Start final polish.

2

weeks after Session 5

to submit everything

2 weeks after Session 5 — submit your GitHub repo AND your peer review together. No separate deadlines.

Referral to partner companies

Top projects are referred to our partner companies — the same referral route already offered to community moderators.

HOW YOU QUALIFY

- Complete the end-to-end project and earn the project certificate
- Score highly on the 10-point rubric — same criteria for both tracks
- Ship a repository a reviewer can run without asking a question
- Complete your peer review on time

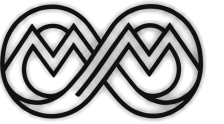
Attendance certificate alone does not qualify

WHAT YOU GET

Your project put in front of hiring partners, by name.

Referrals go out after Session 5, once peer reviews are in.

At the end



What I need from you!

